**Ans.** 4 equally likely outcomes: $\frac{1}{4}$.

**5.** Running thread · Method [ALG] Procedure: choose a number; square it; add 1. Result for $-3$? Can 0 be obtained? Justify.

**Ans.** For $-3 : (-3)^2 + 1 = 10$. No: $x^2 \geqslant 0$ therefore $x^2 + 1 \geqslant 1 > 0$.

**Sheet 29** Mon. 11 Jan. 2027 Grouped data: histogram and cumulative frequencies

**1.** Warm-up · Flash [STA] Give the width and the midpoint of the class $[20;35[$.

**Ans.** Width 15; midpoint $\frac{20+35}{2} = 27{,}5$.

**2.** Core · Table [LEC] Classes $[0;10[$, $[10;20[$, $[20;30[$ with frequencies 4, 6, 10. Give the relative frequencies then the cumulative relative frequencies.

**Ans.** Total 20; frequencies 20 %, 30 %, 50 %; cumulative 20 %, 50 %, 100 %.

**3.** Core + · Flash [LEC] In a histogram with classes of equal width, two rectangles have heights 3 cm and 6 cm. What is the ratio of their frequencies?

**Ans.** The heights are proportional to the frequencies: 1 to 2.

**4.** Back · Flash [NUM] Write 0.00032 in scientific notation.

**Ans.** $3{,}2 \times 10^{-4}$.

**5.** Running thread · Flash [EQ] Solve in $\mathbb{R} : 2(x+3) = 5x - 9$.

**Ans.** $x = 5$

Pitfall: forgetting to distribute the 2 over the 3.

**Sheet 30** Tue. 12 Jan. 2027 Weighted mean, median class; spreadsheet and Python

**1.** Warm-up · Flash [CAL] Calculate $\frac{2 \times 8 + 3 \times 12}{5}$.

**Ans.** $\frac{52}{5} = 10{,}4$.

**2.** Core · Method [STA] Classes $[0;10[$, $[10;20[$, $[20;30[$ with frequencies 4; 7; 9. Estimate the mean using the class midpoints; give the median class.

**Ans.** Estimated mean $\frac{4 \times 5 + 7 \times 15 + 9 \times 25}{20} = 17.5$; cumulative frequencies 4; 11; 20: the $10^{\text{e}}$ and $11^{\text{e}}$ values are in $[10;20[$ : median class.

**3.** Core + · Reading code [ALG] What does this program calculate and what value is obtained?

```
notes = [12, 15, 9, 14]
s = 0
for n in notes:
    s = s + n
print(s / len(notes))
```

**Ans.** Mean mark: $\frac{50}{4} = 12{,}5$.

**4.** Back · Flash [PRO] A 45,000 Ar item is on sale at $-30\%$. What is the sale price?

**Ans.** $45\,000 \times 0{,}70 = 31{,}500$ Ar.

**5.** Running thread · Prove [LIT] Procedure: choose $n$ ; add 1; square; subtract $n^2$. Show that the result is always an odd number.

**Ans.** $(n+1)^2 - n^2 = 2n + 1$: odd for every integer $n$.

## Algebraic calculation, equations and inequalities

**Sheet 31** Thu. 14 Jan. 2027 Integer powers (positive and negative)

**1.** Warm-up · Flash [PUI] Calculate $2^{-3}$, $10^{-2}$ and $(-3)^2$.

**Ans.** $\frac{1}{8}$ ; 0,01 ; 9.

**2.** Core · Method [PUI] Simplify $\frac{2^3 \times 2^{-5}}{2^{-1}}$.

**Ans.** $\frac{2^{-2}}{2^{-1}} = 2^{-2+1} = 2^{-1} = \frac{1}{2}$.

**3.** Core + · Method [PUI] For $x \neq 0$, write $(x^2)^{-3} \times x^7$ in the form $x^n$.

**Ans.** $x^{-6} \times x^7 = x^1 = x$.

**4.** Back · Flash [STA] Median of the data set 4, 7, 9, 10, 12, 15?

**Ans.** Even count: $\frac{9+10}{2} = 9{,}5$.

**5.** Running thread · Recognise [SOL] Which solid has a disc-shaped base, a curved lateral surface and a single vertex?

**Ans.** A right circular cone.

**Sheet 32** Mon. 18 Jan. 2027 Square roots; $\sqrt{a^2} = |a|$

**1.** Warm-up · Flash [CAL] Calculate $\sqrt{36}$ and $\sqrt{0{,}09}$.

**Ans.** 6 ; 0,3.

**2.** Core · Method [CAL] Write $\sqrt{72}$ in the form $a\sqrt{2}$ and calculate $\sqrt{(-5)^2}$.

**Ans.** $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$ ; $\sqrt{(-5)^2} = |-5| = 5$.

**3.** Core + · True/false [CAL] True or false? Justify. $\sqrt{9+16} = \sqrt{9} + \sqrt{16}$.

**Ans.** False. $\sqrt{25} = 5$ but $3 + 4 = 7$. Only the product "splits": $\sqrt{ab} = \sqrt{a}\sqrt{b}$.

**4.** Back · Flash [PUI] Calculate $(3^2)^3$.

**Ans.** $3^{2\times 3} = 3^6 = 729$.

**5.** Running thread · True/false [DEM] True or false? Justify. The diagonals of a rhombus are perpendicular.

**Ans.** True. Property of the rhombus (its diagonals are also perpendicular bisectors of each other).

**Sheet 33** Tue. 19 Jan. 2027 Algebraic fractions; expanded, simplified or factorised forms

**1.** Warm-up · Flash [CAL] Calculate $\frac{1}{2} + \frac{1}{3}$.

**Ans.** $\frac{5}{6}$.

**2.** Core · Method [LIT] Write as a single fraction $\frac{1}{x} + \frac{1}{x+1}$ (for $x \neq 0$ and $x \neq -1$).

**Ans.** $\frac{x+1+x}{x(x+1)} = \frac{2x+1}{x(x+1)}$.

**3.** Core + · Method [DEV] Factorise $x^2 - 4$ then simplify $\frac{x^2-4}{x-2}$ for $x \neq 2$.

**Ans.** $x^2 - 4 = (x-2)(x+2)$ ; therefore $\frac{x^2-4}{x-2} = x + 2$.

The choice of form depends on the aim: factorised to simplify, expanded to compare.

**4.** Back · Flash [SUB] Calculate $2a - 3b$ for $a = 2$ and $b = -1$.

**Ans.** $4 + 3 = 7$.

**5.** Running thread · Flash [REP] ABCDEFGH is a cuboid with $AB = 4$, $AD = 3$, $AE = 2$. In the coordinate system $(A; B, D, E)$, what are the coordinates of $G$ ?

**Ans.** $G(4; 3; 2)$ : $G$ is the vertex opposite $A$.

A third coordinate appears in space.

**Sheet 34** Thu. 21 Jan. 2027 Inequalities: sum, multiplication by a real number

**1.** Warm-up · Flash [CMP] Complete with $<$ or $>$ : $-3 \ldots -2$ ; $\frac{1}{3} \ldots 0{,}3$.

**Ans.** $-3 < -2$ ; $\frac{1}{3} > 0{,}3$ ($0{,}333\ldots > 0{,}3$).

**2.** Core · Method [EQ] We have $a < b$. Compare $-2a$ and $-2b$, then $a+5$ and $b+5$.

**Ans.** $-2a > -2b$ (multiplying by a negative reverses the inequality); $a+5 < b+5$.

**3.** Core + · Method [EQ] Given $2 < x < 3$ and $1 < y < 4$, bound $x+y$ then $x-y$.

**Ans.** $3 < x+y < 7$; for $x-y$ : $-y \in ]-4;-1[$ so $-2 < x-y < 2$.

Pitfall: subtracting inequalities term by term.

**4.** Back · Flash [CAL] Calculate $(-3)\times(-4)+2\times(-5)$.

**Ans.** $12-10=2$.

**5.** Running thread · Problem [DIV] A 120 cm by 84 cm sheet is cut into identical squares, with no waste. What is the largest possible side length?

**Ans.** GCD $(120\,;\,84)=12$ : squares with side 12 cm ($10 \times 7 = 70$ squares).

**Sheet 35** Mon. 25 Jan. 2027 Comparison by difference and quotient

**1.** Warm-up · Flash [CMP] Compare $\frac{2}{3}$ and $\frac{3}{4}$ by studying the sign of the difference.

**Ans.** $\frac{2}{3}-\frac{3}{4}=-\frac{1}{12}<0$ : $\frac{2}{3}<\frac{3}{4}$.

**2.** Core · Method [LIT] Compare $x^2+1$ and $2x$ for every real number $x$.

**Ans.** $x^2+1-2x=(x-1)^2\geqslant 0$ : $x^2+1\geqslant 2x$, with equality if and only if $x=1$.

**3.** Core + · Method [CMP] Compare $\frac{7}{5}$ and $\frac{8}{6}$ by quotient (positive numbers).

**Ans.** $\frac{7/5}{8/6}=\frac{42}{40}>1$, hence $\frac{7}{5}>\frac{8}{6}$.

**4.** Back · Table [LEC] Monthly precipitation (fictitious data).

| Month | Jan. | Feb. | March | Apr. |
|---|---|---|---|---|
| Rain (mm) | 280 | 300 | 240 | 120 |

Which month is the wettest? What is the total over these 4 months?

**Ans.** February (300 mm); 940 mm.

**5.** Running thread · Table → formula [REG] $x$ : $-2, -1, 0, 1$ and $f(x)$ : 4, 1, 0, 1. Is the function affine? Propose a formula.

**Ans.** No (the differences are not constant). $f(x)=x^2$ works.

Test: f $(1)-f(0)=1$ but f $(0)-f(-1)=-1$.

**Sheet 36** Tue. 26 Jan. 2027 Equations and inequalities of degree $1^{er}$ ; x $^2$ = a; zero product

**1.** Warm-up · Flash [EQ] Solve $3 \quad x-5=7$.

**Ans.** $x=4$.

**2.** Core · Method [EQ] Solve $2 \quad x+3\leqslant 7-x$.

**Ans.** $3x\leqslant 4$ : $x\leqslant\frac{4}{3}$ ; $S=\left]-\infty;\frac{4}{3}\right]$.

**3.** Core + · Method [EQ] Solve $(x-2)(2x+6)=0$ then $x^2=7$.

**Ans.** Zero product: $x=2$ or $x=-3$ ; $x^2=7$ : $x=\sqrt{7}$ or $x=-\sqrt{7}$.

**4.** Back · Flash [CMP] Compare 1.4 and $\sqrt{2}$ without a calculator.

**Ans.** $1{,}4^2=1{,}96<2$ so $1.4 \quad <\sqrt{2}$ (the square function is increasing on $[0;+\infty[$).

**5.** Running thread · Flash [GEO] $ABC$ is isosceles at $A$ with $\widehat{BAC}=40°$. What is the measure of $\widehat{ABC}$?

**Ans.** $(180-40)\div 2=70°$.